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SE2A

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Leichtbau



Aerodynamic Design of a Fowler Flap for a Hybrid Laminar Flow Control Airfoil

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Tasks

- Define and modify an existing design methodology for a rapid CFD analysis of airfoils.
- Perform parametric studies of a Fowler flap.
- Obtain a new design and compare key aerodynamic characteristics between the original and the final configurations.

Objective

- Maximize lift for takeoff (15°) and landing (30°) flap settings.

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High-lift devices

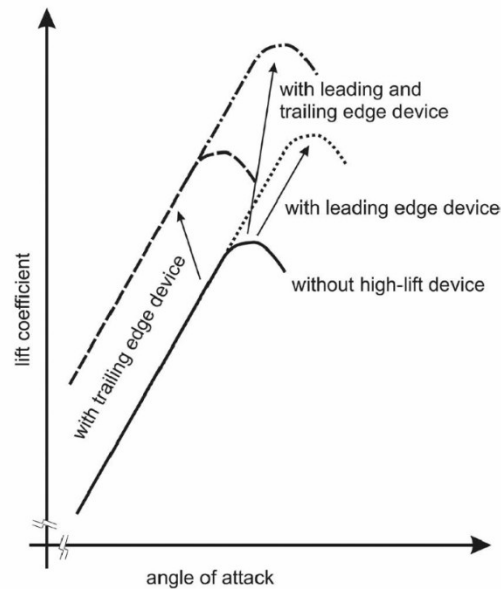
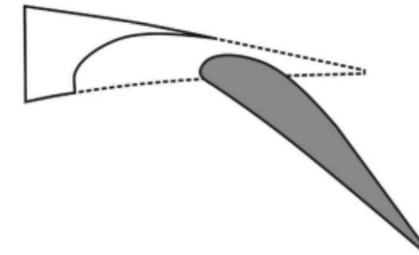


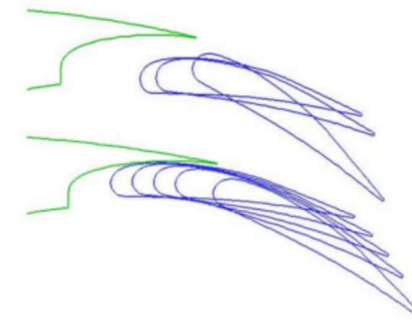
Fig 1: Effect of high-lift devices on the lift coefficient (C_L) and the angle of attack (α) [1].

$$L = \frac{1}{2} \rho V^2 S C_L$$

L = Lift force (N)
 ρ = Air density (kg/m^3)
 V = Freestream velocity (m/s)
 S = Reference area (m^2)
 C_L = Lift coefficient (-)



(a) Principle of a fowler flap.



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(b) Kinematics of a fowler flap.

Fig 2: Mechanism of the Fowler flap [1], [2].

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PyAeroSweep framework

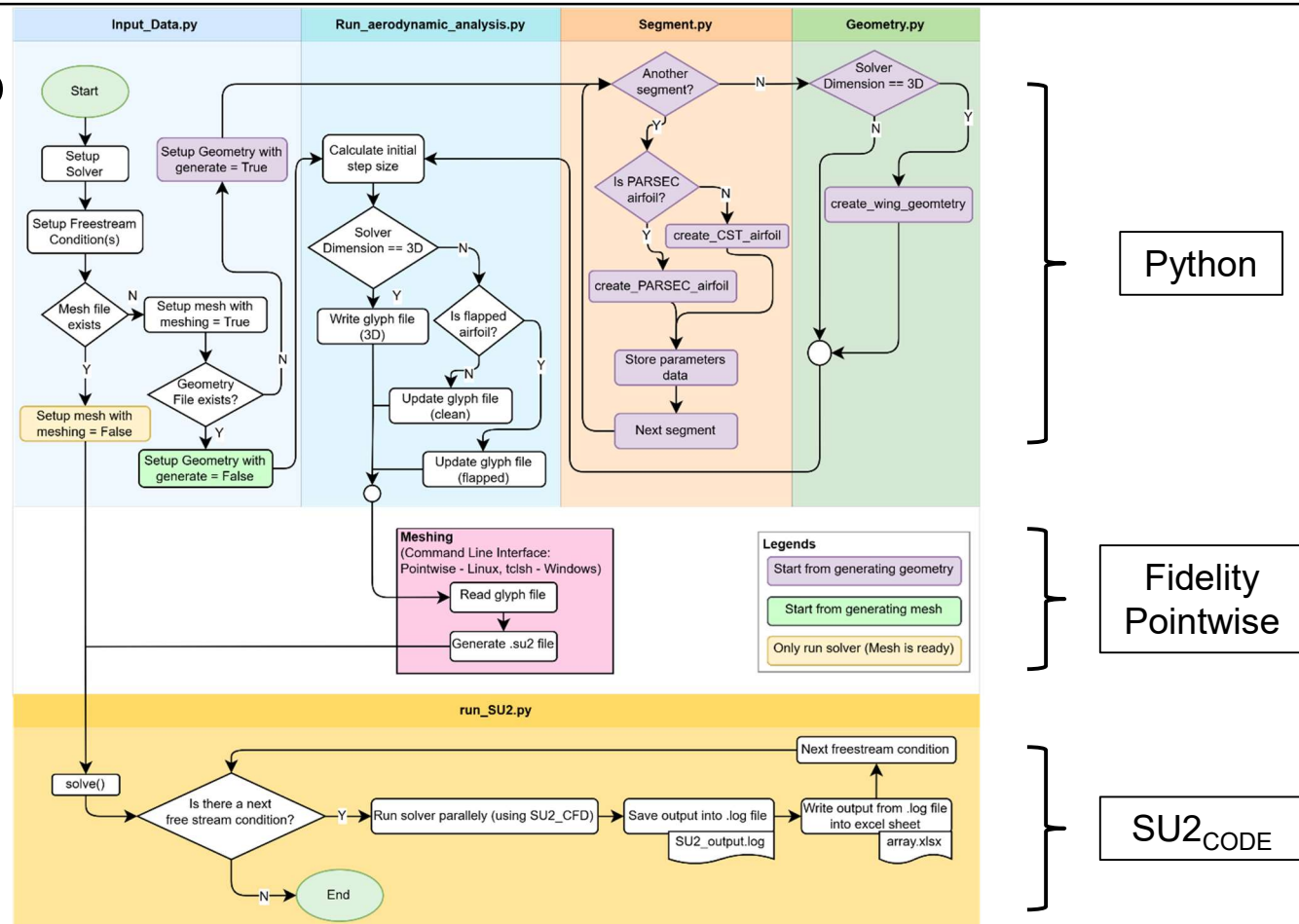
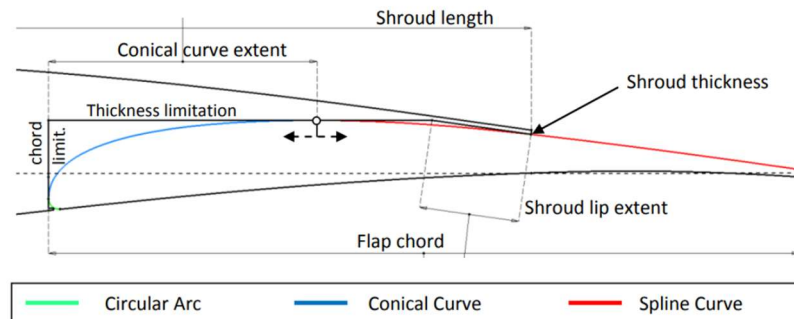
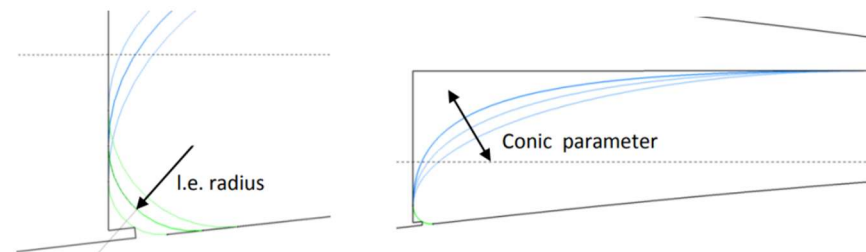


Fig 6: PyAeroSweep Framework.

Geometry definition



(a) Overview: Flap design parameters.



(b) Flap leading edge design parameters.

Fig 3: Key flap geometry definitions [3].

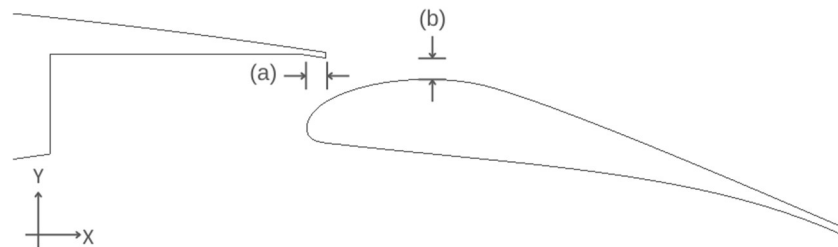


Fig 4: (a) Horizontal overlap x_{gap} and (b) Vertical gap y_{gap} definition.

Geometric parameters – baseline flap

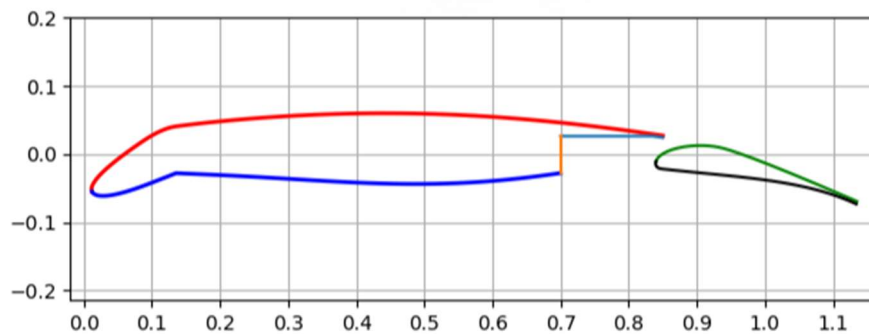


Fig 5: PARSEC airfoil.

15° droop nose
15°/30° flap deflection

(a) Parameters defined with respect to the airfoil chord.

Parameter	Notation	Baseline (ratios)
Flap chord ratio	cf_c	0.3
Shroud chord ratio	csr_c	0.85
Shroud tip thickness	tc_shr_tip	0.003
X-Gap	x_gap	0.01
Y-Gap	y_gap	0.005

(b) Parameters defined with respect to the flap chord.

Parameter	Notation	Baseline (ratios)
Shroud lip extent ratio	clip_ext	0.05
Conical curve extent ratio	ce_c	0.3
Flap leading edge radius	r_le_flap	0.01
Conical parameter	w_conic	0.5
Flap deflection	delta_f	15°/30°

Table 1: Geometric parameters for the baseline flap design.

Mesh generation – Fidelity Pointwise

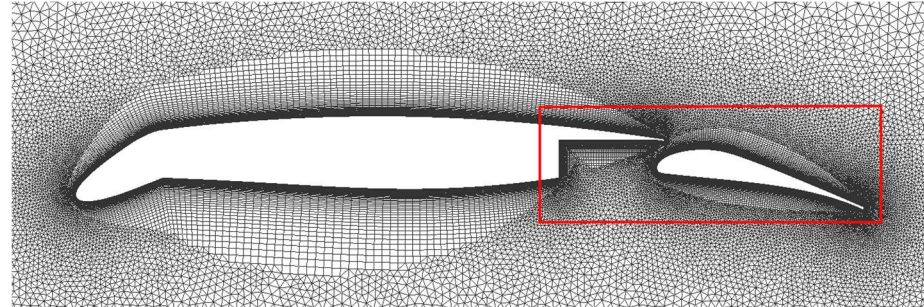


Fig 7: Computational mesh around the HLFC airfoil.

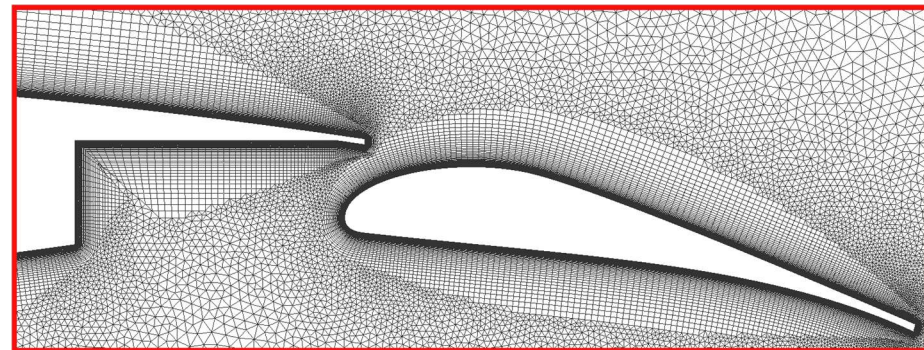


Fig 8: Detailed view of the computational mesh around the Fowler flap.

CFD solver – SU2

- Steady-state Reynolds-Averaged Navier-Stokes (RANS) equations.
 - $k-\omega$ SST (Shear Stress Transport) – Two equation turbulence model.
 - Better handling of adverse pressure gradients and flow separation.
 - Enhanced near-wall resolution.

k : Turbulent kinetic energy (m^2/s^2)
 ω : Specific Rate of Dissipation ($1/\text{s}$)

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Optimization strategy

OAT
One-At-A-Time Optimization

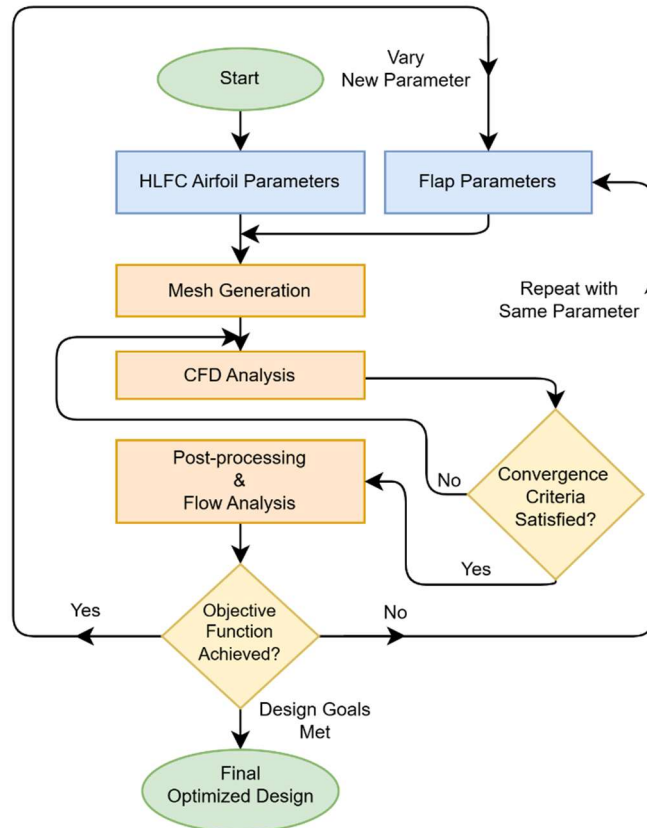


Fig 9: Fowler flap design optimization strategy.

Mesh sensitivity study

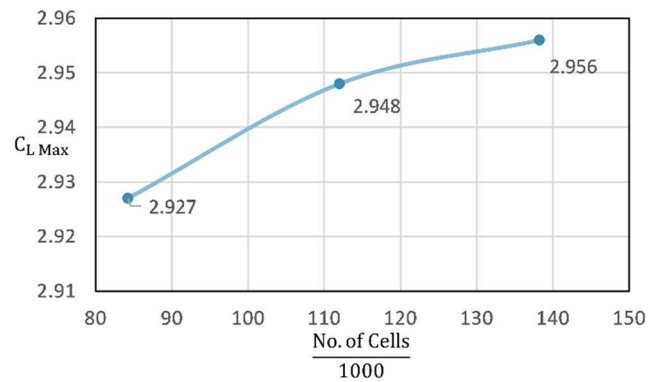
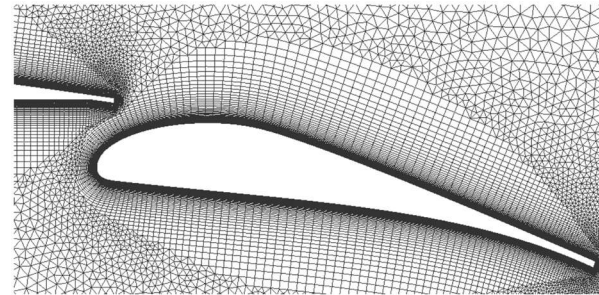


Fig 10: Impact of mesh refinement on maximum lift coefficient ($C_{L_{max}}$).

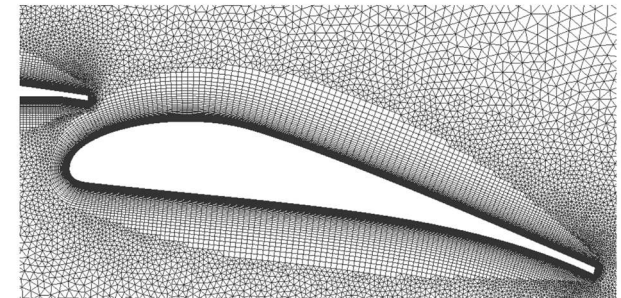
Mesh Type	Cell Count	Connector Scaling	$C_{L_{max}}$
Fine	138,244	+30%	2.956
Baseline	111,992	Baseline	2.948
Coarse	84,225	-30%	2.927

+0.27%
+0.71%

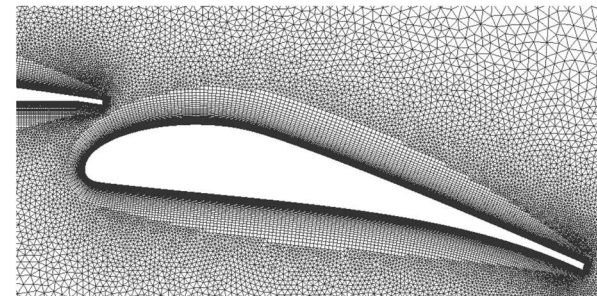
Table 2: Results of the mesh sensitivity study.



(a) Coarse mesh.



(b) Baseline mesh.



(c) Fine mesh.

Fig 11: Effect of mesh refinement on grid resolution.

Parameter sensitivity study

Parameter	Value	Units
Mach Number (Ma)	0.21	-
Altitude	0	meters (Sea Level)
Angle of Attack (α)	4, 6, 8, 10, 11, 12, 13, 14, 15	degrees
Reference Length (C) (Airfoil Chord)	3.28	meters

Table 3: Simulation freestream conditions and reference length (airfoil chord).

Table 4: Range of geometric parameters for the to-be optimized flap design.

(a) Parameters defined with respect to the airfoil chord.

Parameter	Notation	Baseline	Range (ratios)
Flap chord ratio	cf_c	0.3	0.2, 0.25, 0.3
Shroud tip thickness	tc_shr_tip	0.003	0.003, 0.005, 0.007
Shroud chord ratio	csr_c	0.85	0.8, 0.85, 0.9
X-Gap	x_gap	0.01	0.0075, 0.0108, 0.0141
Y-Gap	y_gap	0.005	0.0131, 0.0164, 0.0197

(b) Parameters defined with respect to the flap chord.

Parameter	Notation	Baseline	Range (ratios)
Shroud lip extent ratio	clip_ext	0.05	0.0, 0.025, 0.05
Conical curve extent ratio	ce_c	0.3	0.2, 0.3, 0.35
Flap leading edge radius	r_le_flap	0.01	0.01, 0.0135, 0.017
Conical parameter	w_conic	0.5	0.5, 0.75, 1.0
Flap deflection	delta_f	15°/30°	15°, 30°

Effect of variation of x_{gap} & y_{gap} on C_D and C_L varied across α :

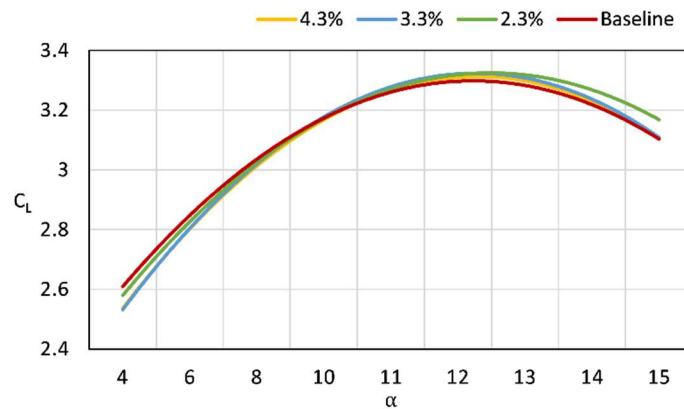


Figure 12: Effect of variation of x_{gap} on C_L vs α , with y_{gap} fixed at 4.0% of the airfoil chord.

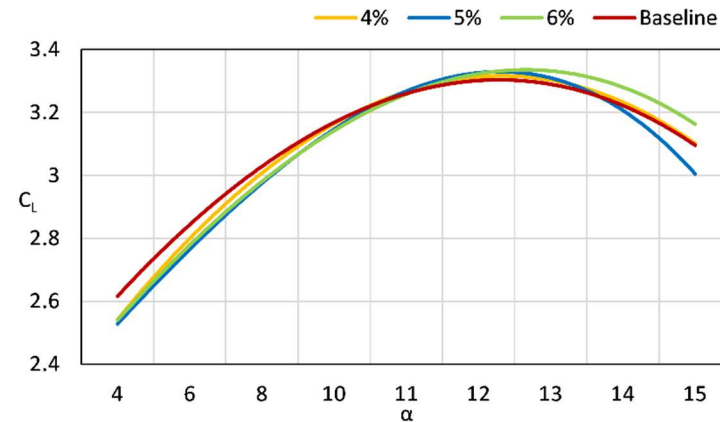


Figure 13: Effect of variation of y_{gap} on C_L vs α , with x_{gap} fixed at 4.3% of the airfoil chord.

Baseline:

x_{gap} 3.0 %
 y_{gap} 1.5 %

Updated x_gap & y_gap

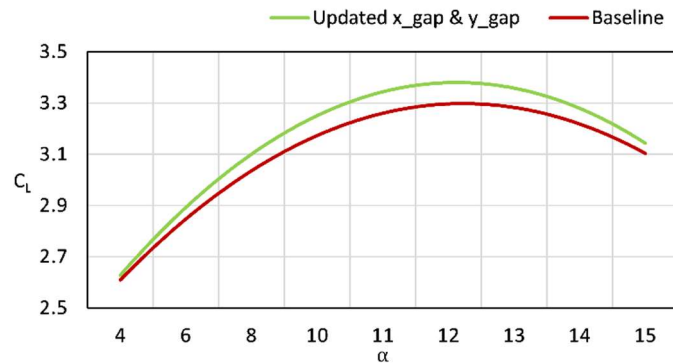


Figure 17: Comparison of baseline and updated configuration (x_gap at 2.3% and y_gap 6.0%).

2.43 % increase in $C_{L_{max}}$

30° flap deflection
4° Angle of attack

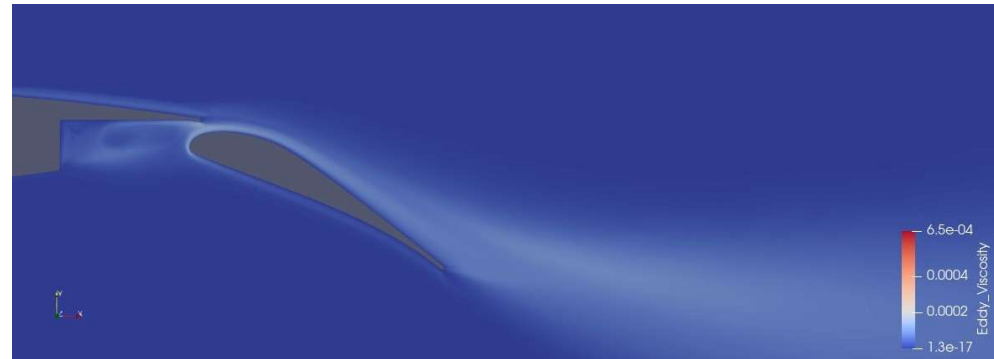


Fig 18: Baseline configuration (x_gap at 3.0% & y_gap at 1.5% of the airfoil chord).

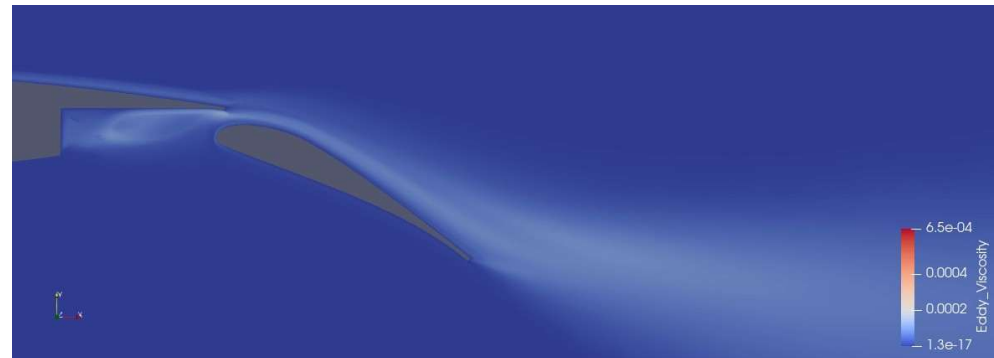


Fig 19: Updated configuration (x_gap at 2.3% & y_gap at 6.0% of the airfoil chord).

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Results

(a) Parameters defined with respect to the airfoil chord.

Parameter	Notation	Baseline	Optimized
Flap chord ratio	cf_c	0.3	0.3
Shroud tip thickness	tc_shr_tip	0.003	0.003
Shroud chord ratio	csr_c	0.85	0.90
X-Gap	x_gap	0.01	0.0075
Y-Gap	y_gap	0.005	0.0197

(b) Parameters defined with respect to the flap chord.

Parameter	Notation	Baseline	Optimized
Shroud lip extent ratio	clip_ext	0.05	0.05
Conical curve extent ratio	ce_c	0.3	0.375
Flap leading edge radius	r_le_flap	0.01	0.0135
Conical parameter	w_conic	0.5	0.75
Flap deflection	delta_f	15°, 30°	15°, 30°

Table 5: Baseline and optimized flap design parameters (in ratios).

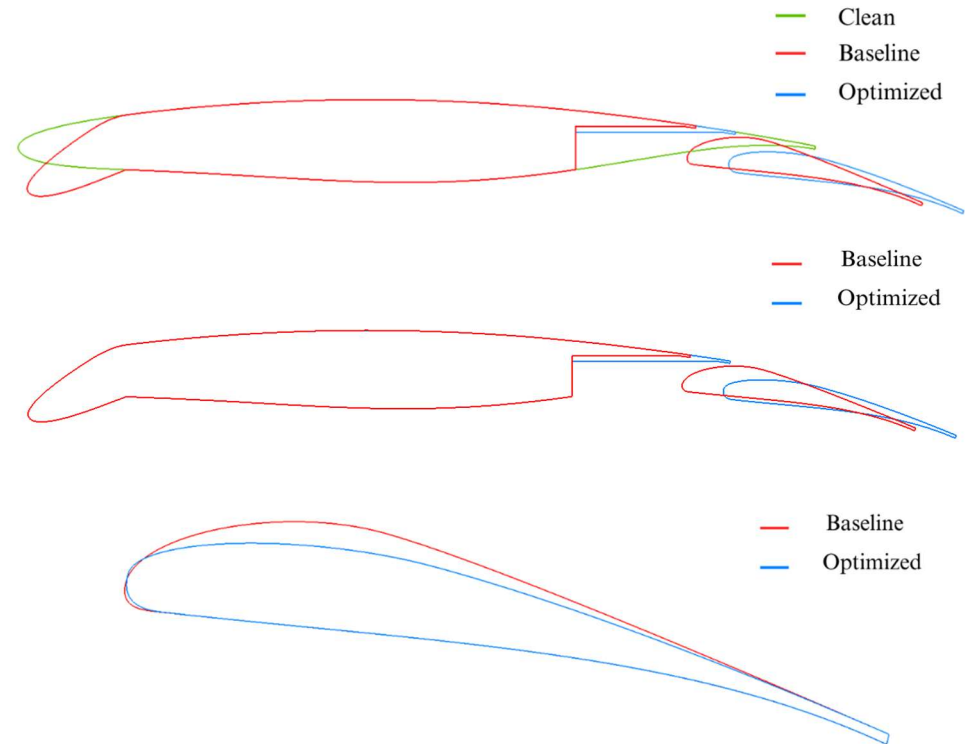


Fig 20: Comparison of baseline and optimized airfoil and flap geometric profiles (all parameters optimized).

Aerodynamic performance at 30-degree flap deflection

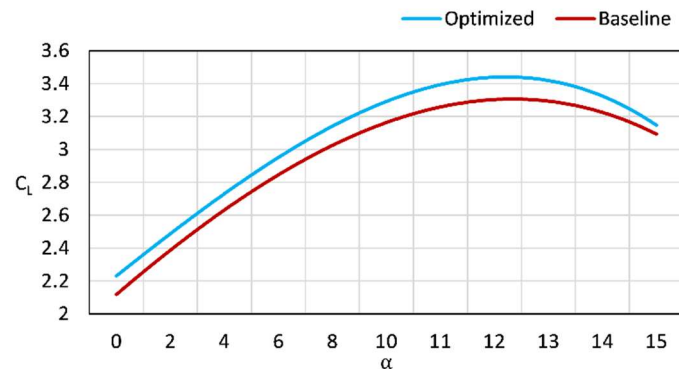


Figure 21: Comparison of baseline and optimized configuration (C_L vs α).

4.56 % increase in
 C_{Lmax}

30° flap deflection
4° Angle of attack

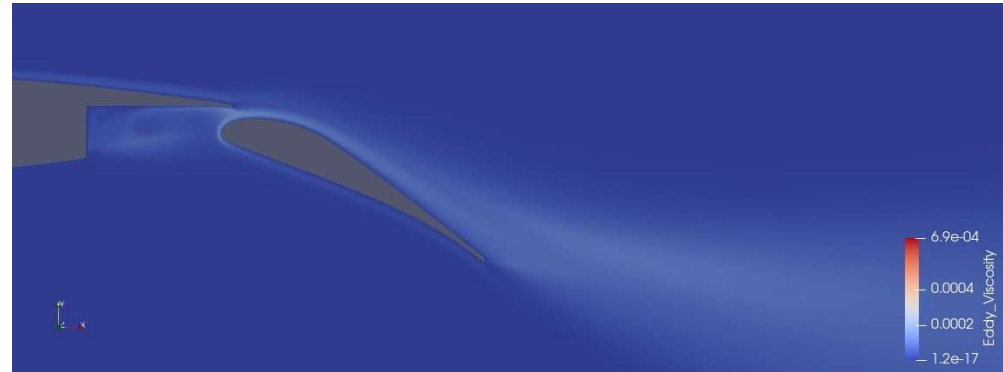


Fig 22: Eddy viscosity: Baseline configuration.

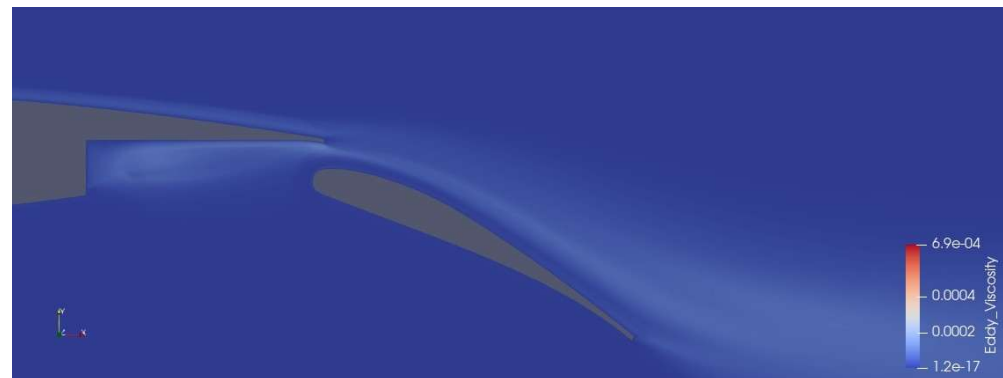


Fig 23: Eddy viscosity: Optimized configuration (all parameters optimized).

Aerodynamic performance at 30-degree flap deflection

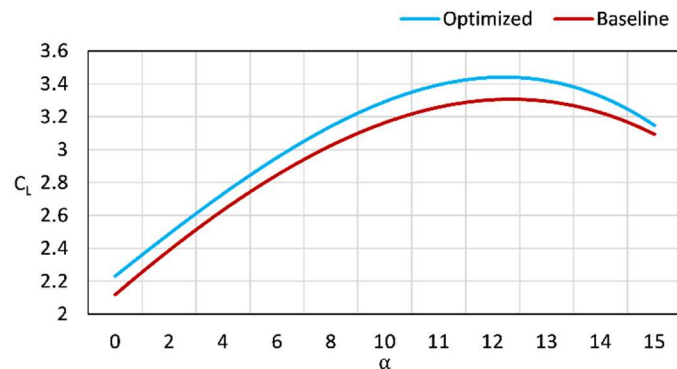


Figure 21: Comparison of baseline and optimized configuration (C_L vs α).

4.56 % increase in
 C_{Lmax}

30° flap deflection
13° Angle of attack

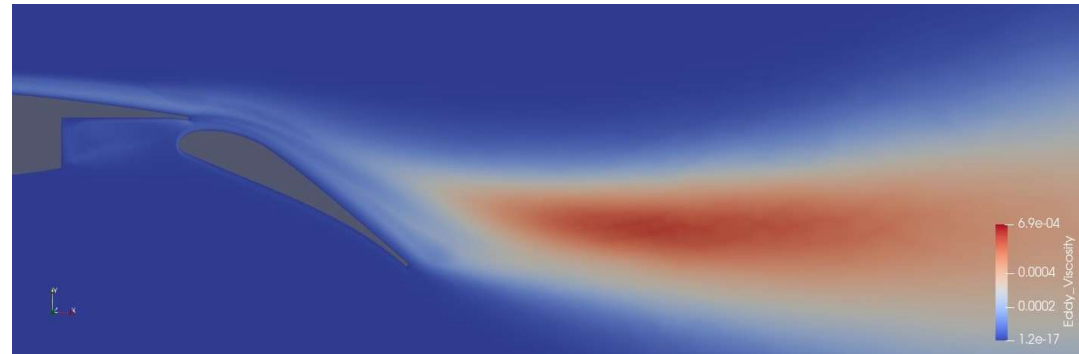


Fig 25: Eddy viscosity: Baseline configuration.

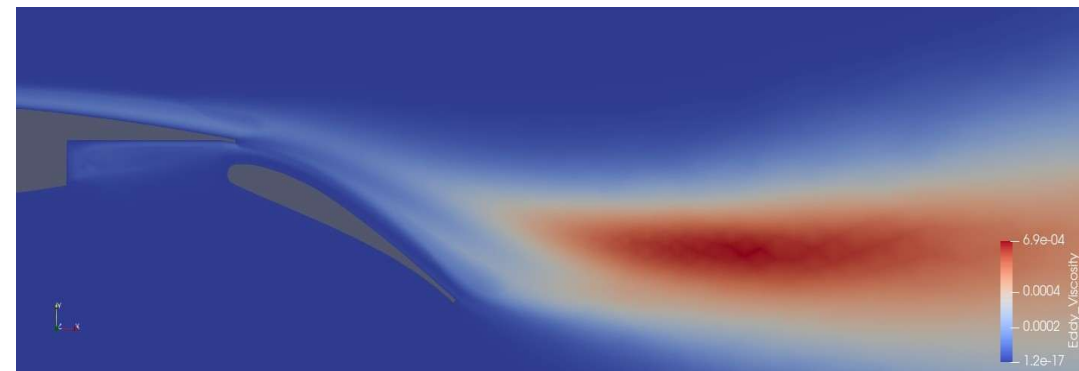


Fig 26: Eddy viscosity Optimized configuration (all parameters optimized).

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Summary

- 1000+ CFD runs, manually optimized and post-processed.
- 10 parameters varied.
- Tested for:
 - 15° flap deflection.
 - 30° flap deflection.

Flap Deflection	Baseline $C_{L_{\max}}$	Optimized $C_{L_{\max}}$	Percentage Improvement
15 degrees	2.90	3.02	4.14%
30 degrees	3.29	3.44	4.56%

Table 6: Summary of maximum lift coefficient ($C_{L_{\max}}$) improvements.

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Future work

- Unsteady Reynolds-Averaged Navier-Stokes (URANS).
- Implementing:
 - Surrogate-Based Optimization.
 - Automatic Optimization Algorithms.

References

- [1] Jochen Wild. *High-lift aerodynamics*. CRC Press, 2022
- [2] Henning Strüber. 'The aerodynamic design of the A350 XWB-900 high lift system'. In: *29th international congress of the aeronautical sciences*. International Council of the Aeronautical Sciences. 2014.
- [3] Mauro Minervino, Pierluigi Iannelli and D. Quagliarella. '3D Flap Design using Navier-Stokes Equations and Evolutionary Optimization Techniques on an Industrial Platform'. In: Sept. 2011. doi: 10.13140/2.1.2211.0083.



Thank you for listening.

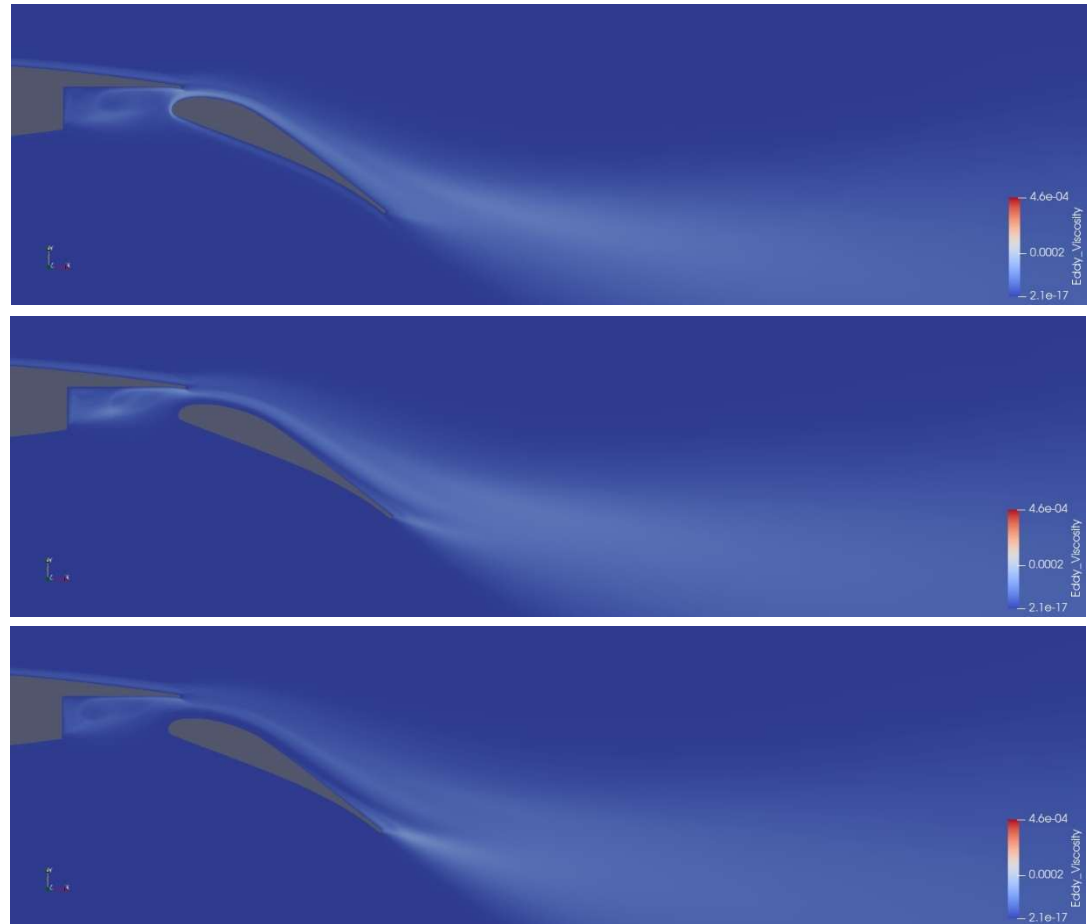
Eddy Viscosity

Fig 14: Baseline configuration
(x_gap at 3.0% & y_gap at 1.5% of
the airfoil chord).

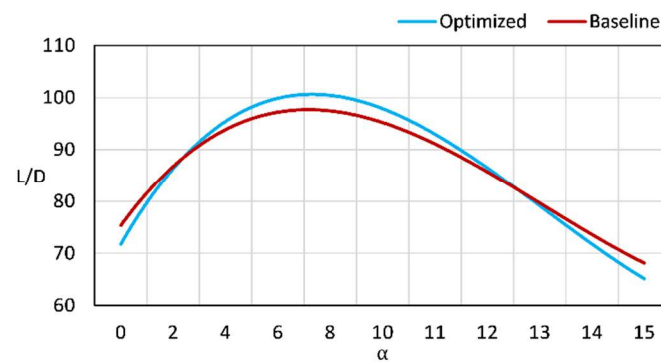
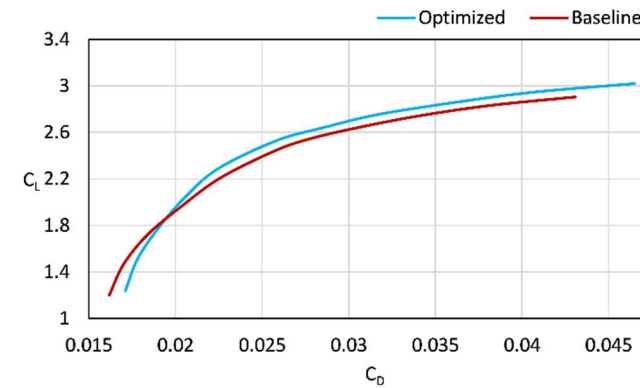
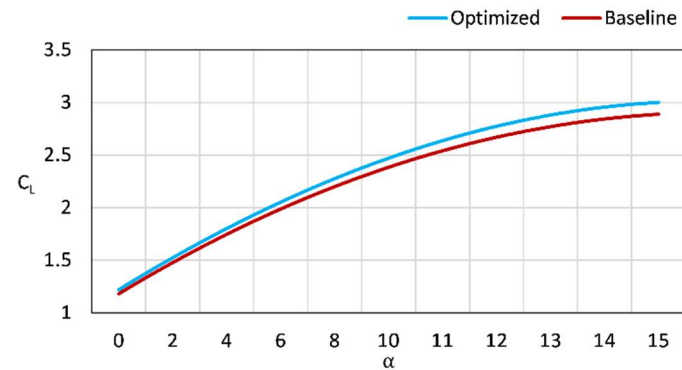
Fig 15: Optimized configuration
(x_gap at 2.3% & y_gap at 4.0% of
the airfoil chord).

Fig 16: Optimized configuration
(x_gap at 4.3% & y_gap at 6.0% of
the airfoil chord).

30° flap deflection
4° Angle of attack



Aerodynamic performance at 15-degree flap deflection



Aerodynamic performance at 30-degree flap deflection

